

Subspaces of $\mathbb{R}^n$

A **subspace** of $\mathbb{R}^n$ is any set H in $\mathbb{R}^n$ that has three properties:

- a. The zero vector is in H .
- b. For each $\mathbf{u}$ and $\mathbf{v}$ in H , the sum $\mathbf{u} + \mathbf{v}$ is in H .
- c. For each $\mathbf{u}$ in H and each scalar c , the vector $c\mathbf{u}$ is in H .

EXAMPLE 1 If $\mathbf{v}_1$ and $\mathbf{v}_2$ are in $\mathbb{R}^n$ and $H = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$, then H is a subspace of $\mathbb{R}^n$. To verify this statement, note that the zero vector is in H (because $0\mathbf{v}_1 + 0\mathbf{v}_2$ is a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$). Now take two arbitrary vectors in H , say,

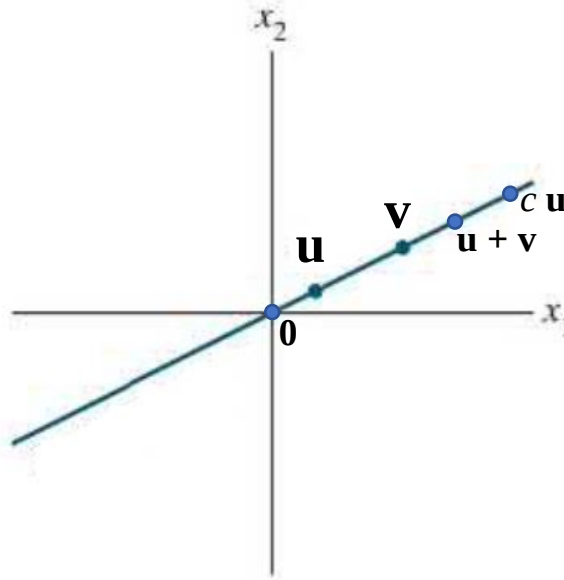
$$\mathbf{u} = s_1\mathbf{v}_1 + s_2\mathbf{v}_2 \quad \text{and} \quad \mathbf{v} = t_1\mathbf{v}_1 + t_2\mathbf{v}_2$$

Then

$$\mathbf{u} + \mathbf{v} = (s_1 + t_1)\mathbf{v}_1 + (s_2 + t_2)\mathbf{v}_2$$

which shows that $\mathbf{u} + \mathbf{v}$ is a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$ and hence is in H . Also, for any scalar c , the vector $c\mathbf{u}$ is in H , because $c\mathbf{u} = c(s_1\mathbf{v}_1 + s_2\mathbf{v}_2) = (cs_1)\mathbf{v}_1 + (cs_2)\mathbf{v}_2$. 

Example A line through the origin is another example of a subspace



All three properties exist:

- includes the zero vector (the origin)
- closed under addition
- closed under scalar multiplication

EXAMPLE 2 A line L *not* through the origin is *not* a subspace, because it does not contain the origin, as required. Also, Figure 2 shows that L is not closed under addition or scalar multiplication. ■

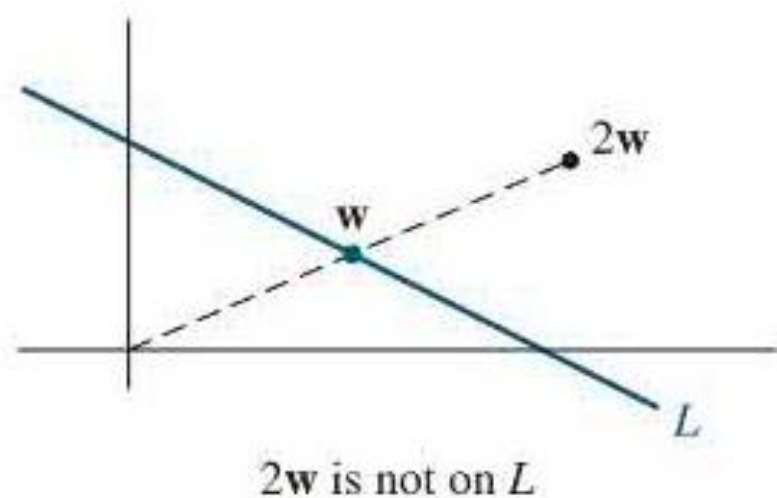
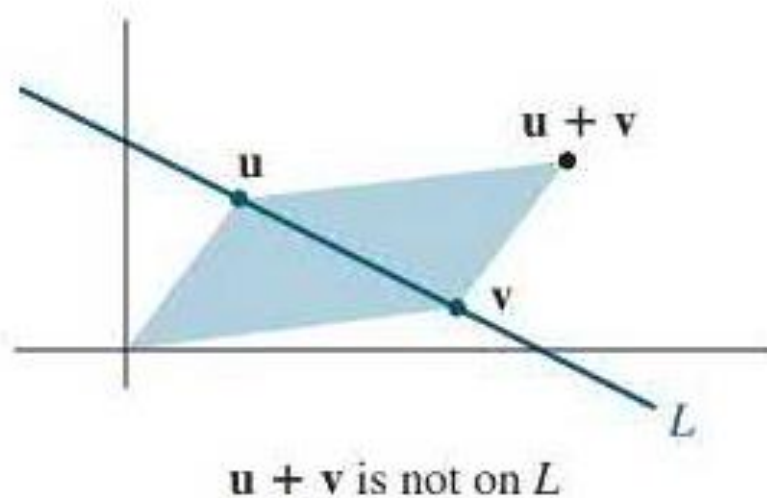


FIGURE 2

EXAMPLE 3 For $\mathbf{v}_1, \dots, \mathbf{v}_p$ in $\mathbb{R}^n$, the set of all linear combinations of $\mathbf{v}_1, \dots, \mathbf{v}_p$ is a subspace of $\mathbb{R}^n$. The verification of this statement is similar to the argument given in Example 1. We shall now refer to $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ as **the subspace spanned** (or **generated**) by $\mathbf{v}_1, \dots, \mathbf{v}_p$. ■

Example The set containing only the **zero vector** is a subspace

All three properties exist:

- includes the zero vector (the origin)
- closed under addition
- closed under scalar multiplication

Example $\mathbb{R}^n$ is a subspace of itself

Column Space and Null Space of a Matrix

The **column space** of a matrix A is the set $\text{Col } A$ of all linear combinations of the columns of A .

In other words
The **column space** is the same as the **span** of the columns of the matrix

To determine if a vector $\mathbf{b}$ is in column space of matrix A ,
Check if equation $A\mathbf{x} = \mathbf{b}$ is consistent

EXAMPLE 4 Let $A = \begin{bmatrix} 1 & -3 & -4 \\ -4 & 6 & -2 \\ -3 & 7 & 6 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 3 \\ 3 \\ -4 \end{bmatrix}$. Determine whether $\mathbf{b}$ is in the column space of A .

SOLUTION The vector $\mathbf{b}$ is a linear combination of the columns of A if and only if $\mathbf{b}$ can be written as $A\mathbf{x}$ for some $\mathbf{x}$, that is, if and only if the equation $A\mathbf{x} = \mathbf{b}$ has a solution. Row reducing the augmented matrix $[A \ \mathbf{b}]$,

$$\begin{bmatrix} 1 & -3 & -4 & 3 \\ -4 & 6 & -2 & 3 \\ -3 & 7 & 6 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & -4 & 3 \\ 0 & -6 & -18 & 15 \\ 0 & -2 & -6 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & -4 & 3 \\ 0 & -6 & -18 & 15 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

we conclude that $A\mathbf{x} = \mathbf{b}$ is consistent and $\mathbf{b}$ is in $\text{Col } A$. ■

The **null space** of a matrix A is the set $\text{Nul } A$ of all solutions of the homogeneous equation $A\mathbf{x} = \mathbf{0}$.

Basis for a Subspace

A **basis** for a subspace H of $\mathbb{R}^n$ is a linearly independent set in H that spans H .

Two conditions must be satisfied in the set of vectors to be a basis of a subspace:

1. The vectors are linearly independent
2. The vectors span the subspace.

The columns of the identity matrix $\{e_1, \dots, e_n\}$ is called **the standard basis for $\mathbb{R}^n$** because they are linearly independent and span $\mathbb{R}^n$

The pivot columns of a matrix A form a basis for the column space of A .

To find the basis of the null space of matrix A ,

1. Solve the homogenous equation $A\mathbf{x} = \mathbf{0}$
2. Write the solution in parametric vector form
3. Take the vectors of $\mathbf{x}$ as the basis

To find the basis of the column space of matrix A ,

1. Reduce to Reduced Echelon Form
2. Take the pivot columns **from the original matrix** (not from the reduced echelon form) as the basis

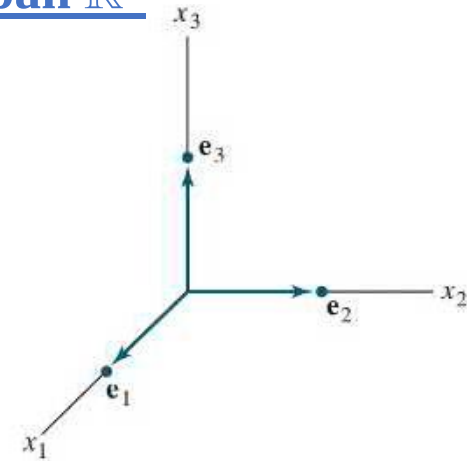


FIGURE 3
The standard basis for $\mathbb{R}^3$.

EXAMPLE 6 Find a basis for the null space of the matrix

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

SOLUTION First, write the solution of $A\mathbf{x} = \mathbf{0}$ in parametric vector form:

$$[A \quad \mathbf{0}] \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{array}{rcl} x_1 - 2x_2 & - & x_4 + 3x_5 = 0 \\ & & x_3 + 2x_4 - 2x_5 = 0 \\ & & 0 = 0 \end{array}$$

The general solution is $x_1 = 2x_2 + x_4 - 3x_5$, $x_3 = -2x_4 + 2x_5$, with x_2 , x_4 , and x_5 free.

$$\begin{aligned} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} &= \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} \\ &\quad \quad \quad \uparrow \quad \quad \quad \uparrow \quad \quad \quad \uparrow \\ &\quad \quad \quad \mathbf{u} \quad \quad \quad \mathbf{v} \quad \quad \quad \mathbf{w} \\ &= x_2 \mathbf{u} + x_4 \mathbf{v} + x_5 \mathbf{w} \end{aligned} \tag{1}$$

Example Find a basis for the column space of the matrix

$$A = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_5] = \begin{bmatrix} 1 & 3 & 3 & 2 & -9 \\ -2 & -2 & 2 & -8 & 2 \\ 2 & 3 & 0 & 7 & 1 \\ 3 & 4 & -1 & 11 & -8 \end{bmatrix}$$

Find the reduced echelon form

$$B = \begin{bmatrix} 1 & 0 & -3 & 5 & 0 \\ 0 & 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

the pivot columns of A are columns 1, 2, and 5.

(if required) Rewrite any nonpivot column as a linear combination of the pivot columns

$$\mathbf{b}_3 = -3\mathbf{b}_1 + 2\mathbf{b}_2 \quad \rightarrow \quad \mathbf{a}_3 = -3\mathbf{a}_1 + 2\mathbf{a}_2$$

$$\mathbf{b}_4 = 5\mathbf{b}_1 - \mathbf{b}_2 \quad \rightarrow \quad \mathbf{a}_4 = 5\mathbf{a}_1 - \mathbf{a}_2$$

Warning:

Use the pivot columns of A (the original matrix) not B (the reduced echelon form)

Therefore, the set $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_5\}$ is a basis for the column space of A (Col A)

Dimension and Rank

Coordinate Systems

Suppose the set $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_p\}$ is a basis for a subspace H . For each $\mathbf{x}$ in H , the **coordinates of $\mathbf{x}$ relative to the basis $\mathcal{B}$** are the weights $c_1, \dots, c_p$ such that $\mathbf{x} = c_1 \mathbf{b}_1 + \dots + c_p \mathbf{b}_p$, and the vector in $\mathbb{R}^p$

$$[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_p \end{bmatrix}$$

is called the **coordinate vector of $\mathbf{x}$ (relative to $\mathcal{B}$)** or the **$\mathcal{B}$ -coordinate vector of $\mathbf{x}$** .¹

EXAMPLE 1 Let $\mathbf{v}_1 = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$, and $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$. Then $\mathcal{B}$ is a basis for $H = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$ because $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly independent. Determine if $\mathbf{x}$ is in H , and if it is, find the coordinate vector of $\mathbf{x}$ relative to $\mathcal{B}$.

SOLUTION If $\mathbf{x}$ is in H , then the following vector equation is consistent:

$$c_1 \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$$

The scalars c_1 and c_2 , if they exist, are the $\mathcal{B}$ -coordinates of $\mathbf{x}$. Row operations show that

$$\begin{bmatrix} 3 & -1 & 3 \\ 6 & 0 & 12 \\ 2 & 1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus $c_1 = 2$, $c_2 = 3$, and $[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$. The basis $\mathcal{B}$ determines a “coordinate system” on H , which can be visualized by the grid shown in Figure 1. ■

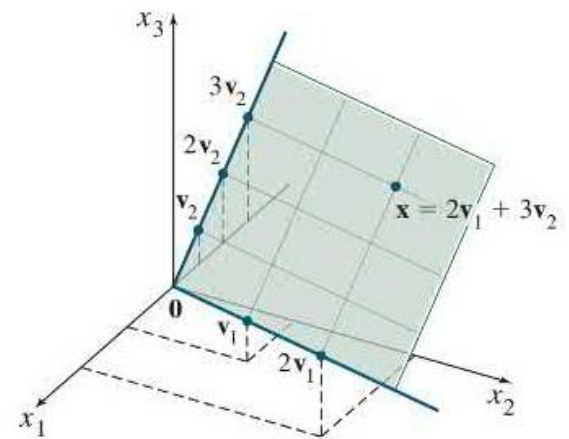


FIGURE 1 A coordinate system on a plane H in $\mathbb{R}^3$.

The Dimension of a Subspace

The **dimension** of a nonzero subspace H , denoted by $\dim H$, is the number of vectors in any basis for H . The dimension of the zero subspace $\{\mathbf{0}\}$ is defined to be zero.²

For the null space, the dimension is simply the number of free variables of the equation $A \mathbf{x} = \mathbf{0}$

The **rank** of a matrix A , denoted by $\text{rank } A$, is the dimension of the column space of A .

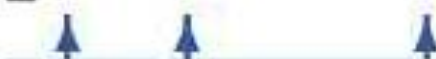
In other words, the number of pivot columns

EXAMPLE 3 Determine the rank of the matrix

$$A = \begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 4 & 7 & -4 & -3 & 9 \\ 6 & 9 & -5 & 2 & 4 \\ 0 & -9 & 6 & 5 & -6 \end{bmatrix}$$

SOLUTION Reduce A to echelon form:

$$A \sim \begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 0 & -6 & 4 & 14 & -20 \\ 0 & -9 & 6 & 5 & -6 \end{bmatrix} \sim \dots \sim \begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 0 & 0 & 0 & 4 & -6 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Pivot columns 

The matrix A has 3 pivot columns, so $\text{rank } A = 3$. 

The Rank Theorem

If a matrix A has n columns, then $\text{rank } A + \dim \text{Nul } A = n$.

The Basis Theorem

Let H be a p -dimensional subspace of $\mathbb{R}^n$. Any linearly independent set of exactly p elements in H is automatically a basis for H . Also, any set of p elements of H that spans H is automatically a basis for H .

The Invertible Matrix Theorem (continued)

Let A be an $n \times n$ matrix. Then the following statements are each equivalent to the statement that A is an invertible matrix.

- m. The columns of A form a basis of $\mathbb{R}^n$.
- n. $\text{Col } A = \mathbb{R}^n$
- o. $\text{rank } A = n$
- p. $\dim \text{Nul } A = 0$
- q. $\text{Nul } A = \{\mathbf{0}\}$

To find the basis of a column space of matrix A :

1. Find the equivalent matrix in Echelon Form or Reduced Echelon Form
2. Take the pivot columns **from the original matrix** (not from the equivalent matrix)

To find the dimension of the column space of matrix A :

Find the number of pivots

To find the rank of matrix A :

Find the number of pivots

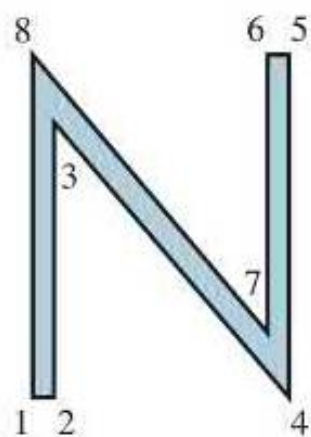
To find the basis of the null space of matrix A ,

1. Solve the homogenous equation $A \mathbf{x} = \mathbf{0}$
2. Write the solution in parametric vector form
3. Take the vectors as the basis

To find the dimension of the null space of matrix A :

Find the number of free variables of the equation $A \mathbf{x} = \mathbf{0}$

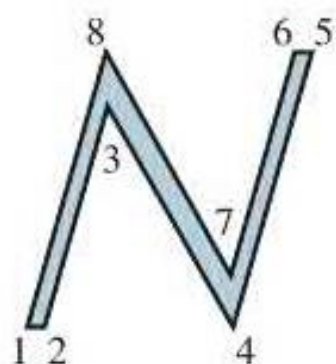
Applications to Computer Graphics



EXAMPLE 1 The capital letter N in Figure 1 is determined by eight points, or *vertices*. The coordinates of the points can be stored in a data matrix, D .

$$\begin{array}{l} \text{Vertex:} \\ \begin{array}{l} x\text{-coordinate} \\ y\text{-coordinate} \end{array} \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 0 & .5 & .5 & 6 & 6 & 5.5 & 5.5 & 0 \\ 0 & 0 & 6.42 & 0 & 8 & 8 & 1.58 & 8 \end{bmatrix} = D \end{array}$$

In addition to D , it is necessary to specify which vertices are connected by lines, but we omit this detail. ■

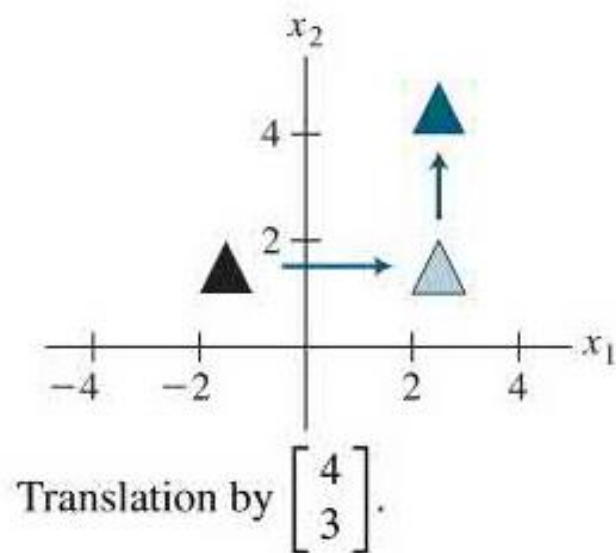


EXAMPLE 2 Given $A = \begin{bmatrix} 1 & .25 \\ 0 & 1 \end{bmatrix}$, describe the effect of the shear transformation $\mathbf{x} \mapsto A\mathbf{x}$ on the letter N in Example 1.

SOLUTION By definition of matrix multiplication, the columns of the product AD contain the images of the vertices of the letter N.

$$AD = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 0 & .5 & 2.105 & 6 & 8 & 7.5 & 5.895 & 2 \\ 0 & 0 & 6.420 & 0 & 8 & 8 & 1.580 & 8 \end{bmatrix}$$

The transformed vertices are plotted in Figure 2, along with connecting line segments that correspond to those in the original figure. ■



Homogeneous Coordinates

Each point (x, y) in $\mathbb{R}^2$ can be identified with the point $(x, y, 1)$ on the plane in $\mathbb{R}^3$ that lies one unit above the xy -plane. We say that (x, y) has *homogeneous coordinates* $(x, y, 1)$. For instance, the point $(0, 0)$ has homogeneous coordinates $(0, 0, 1)$. Homogeneous coordinates for points are not added or multiplied by scalars, but they can be transformed via multiplication by 3×3 matrices.

EXAMPLE 4 A translation of the form $(x, y) \mapsto (x + h, y + k)$ is written in homogeneous coordinates as $(x, y, 1) \mapsto (x + h, y + k, 1)$. This transformation can be computed via matrix multiplication:

$$\begin{bmatrix} 1 & 0 & h \\ 0 & 1 & k \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x + h \\ y + k \\ 1 \end{bmatrix}$$



EXAMPLE 5 Any linear transformation on $\mathbb{R}^2$ is represented with respect to homogeneous coordinates by a partitioned matrix of the form $\begin{bmatrix} A & 0 \\ 0 & 1 \end{bmatrix}$, where A is a 2×2 matrix. Typical examples are

$$\begin{bmatrix} \cos \varphi & -\sin \varphi & 0 \\ \sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

Counterclockwise
rotation about the
origin, angle φ

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

Reflection
through $y = x$

$$\begin{bmatrix} s & 0 & 0 \\ 0 & t & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Scale x by s
and y by t



Composite Transformations

The movement of a figure on a computer screen often requires two or more basic transformations. The composition of such transformations corresponds to matrix multiplication when homogeneous coordinates are used.

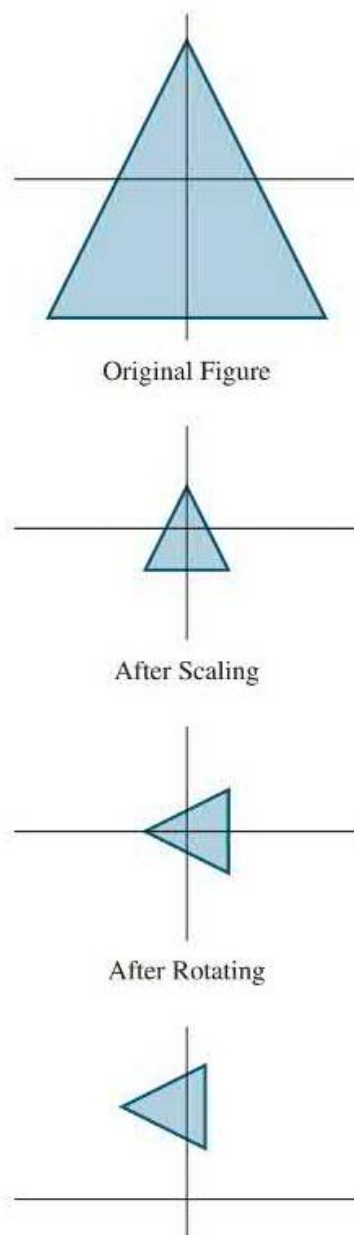
EXAMPLE 6 Find the 3×3 matrix that corresponds to the composite transformation of a scaling by .3, a rotation of 90° about the origin, and finally a translation that adds $(-.5, 2)$ to each point of a figure.

SOLUTION If $\varphi = \pi/2$, then $\sin \varphi = 1$ and $\cos \varphi = 0$. From Examples 4 and 5, we have

$$\begin{aligned} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} &\xrightarrow{\text{Scale}} \begin{bmatrix} .3 & 0 & 0 \\ 0 & .3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \\ &\xrightarrow{\text{Rotate}} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} .3 & 0 & 0 \\ 0 & .3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \\ &\xrightarrow{\text{Translate}} \begin{bmatrix} 1 & 0 & -.5 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} .3 & 0 & 0 \\ 0 & .3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \end{aligned}$$

The matrix for the composite transformation is

$$\begin{aligned} \begin{bmatrix} 1 & 0 & -.5 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} .3 & 0 & 0 \\ 0 & .3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ = \begin{bmatrix} 0 & -1 & -.5 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} .3 & 0 & 0 \\ 0 & .3 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -.3 & -.5 \\ .3 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix} \quad \blacksquare \end{aligned}$$



Homogeneous 3D Coordinates

In general, (X, Y, Z, H) are **homogeneous coordinates**

$$x = \frac{X}{H}, \quad y = \frac{Y}{H}, \quad \text{and} \quad z = \frac{Z}{H} \quad (1)$$

EXAMPLE 7 Give 4×4 matrices for the following transformations:

- Rotation about the y -axis through an angle of 30° . (By convention, a positive angle is the counterclockwise direction when looking toward the origin from the positive half of the axis of rotation—in this case, the y -axis.)
- Translation by the vector $\mathbf{p} = (-6, 4, 5)$.

SOLUTION

- First, construct the 3×3 matrix for the rotation. The vector $\mathbf{e}_1$ rotates down toward the negative z -axis, stopping at $(\cos 30^\circ, 0, -\sin 30^\circ) = (\sqrt{3}/2, 0, -.5)$. The vector $\mathbf{e}_2$ on the y -axis does not move, but $\mathbf{e}_3$ on the z -axis rotates down toward the positive

$$\begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix}$$

x -axis, stopping at $(\sin 30^\circ, 0, \cos 30^\circ) = (.5, 0, \sqrt{3}/2)$. See Figure 5. From Section 1.9, the standard matrix for this rotation is

$$\begin{bmatrix} \sqrt{3}/2 & 0 & .5 \\ 0 & 1 & 0 \\ -.5 & 0 & \sqrt{3}/2 \end{bmatrix}$$

So the rotation matrix for homogeneous coordinates is

$$A = \begin{bmatrix} \sqrt{3}/2 & 0 & .5 & 0 \\ 0 & 1 & 0 & 0 \\ -.5 & 0 & \sqrt{3}/2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- We want $(x, y, z, 1)$ to map to $(x - 6, y + 4, z + 5, 1)$. The matrix that does this is

$$\begin{bmatrix} 1 & 0 & 0 & -6 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Perspective Projections

$$P \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1/d & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ 0 \\ 1 - z/d \end{bmatrix}$$

EXAMPLE 8 Let S be the box with vertices $(3, 1, 5)$, $(5, 1, 5)$, $(5, 0, 5)$, $(3, 0, 5)$, $(3, 1, 4)$, $(5, 1, 4)$, $(5, 0, 4)$, and $(3, 0, 4)$. Find the image of S under the perspective projection with center of projection at $(0, 0, 10)$.

SOLUTION Let P be the projection matrix, and let D be the data matrix for S using homogeneous coordinates. The data matrix for the image of S is

$$PD = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1/10 & 1 \end{bmatrix} \begin{array}{c} \text{Vertex:} \\ \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 5 & 5 & 3 & 3 & 5 & 5 & 3 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 5 & 5 & 5 & 5 & 4 & 4 & 4 & 4 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} \end{array}$$

$$= \begin{bmatrix} 3 & 5 & 5 & 3 & 3 & 5 & 5 & 3 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ .5 & .5 & .5 & .5 & .6 & .6 & .6 & .6 \end{bmatrix}$$

To obtain $\mathbb{R}^3$ coordinates, use equation (1) before Example 7, and divide the top three entries in each column by the corresponding entry in the fourth row:

$$\begin{array}{c} \text{Vertex:} \\ \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 6 & 10 & 10 & 6 & 5 & 8.3 & 8.3 & 5 \\ 2 & 2 & 0 & 0 & 1.7 & 1.7 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

